2022 SCIENCE OLYMPIAD NATIONAL INVITATIONAL

JANUARY 8, 2022

Dynamic Planet C

FROM THE HOSTS OF THE 2022 NATIONAL TOURNAMENT

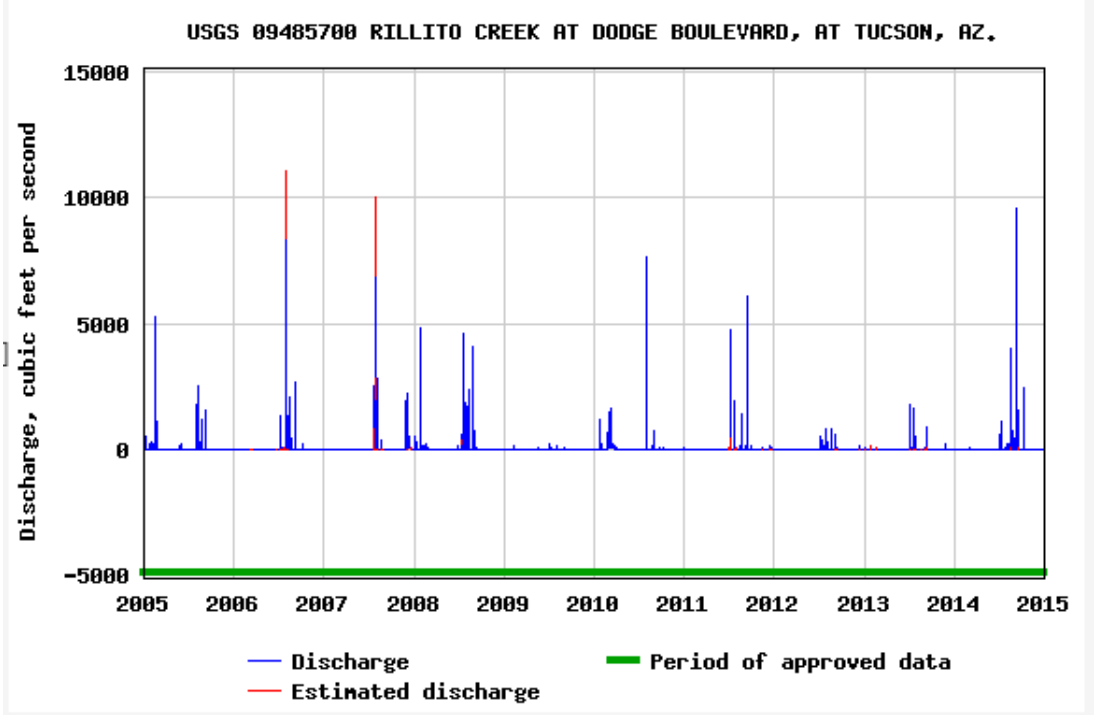
SOUTHERN CALIFORNIA SCIENCE OLYMPIAD

Caltech

SCIENCE OLYMPIAD™

Exploring the World of Science

1. (1 point) The figure below shows a USGS stream gauge record.



Is the streamflow at this gauge

A. Perennial

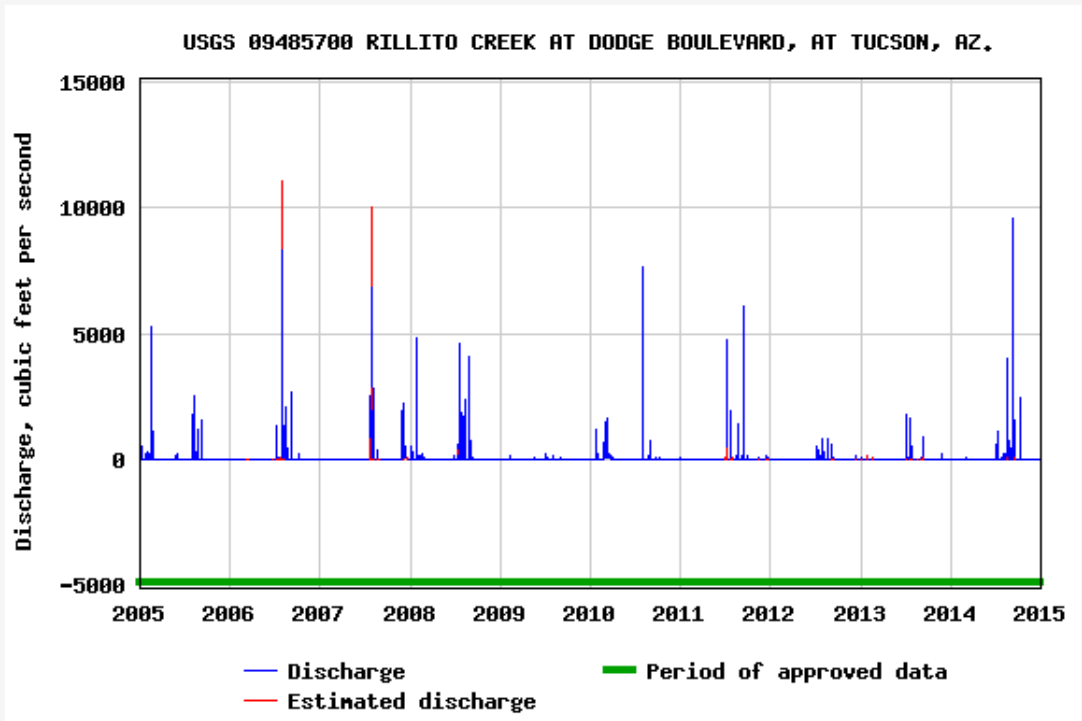
B. Intermittent

C. Tidal

D. Varzea

E. It is impossible to say

2. (2 points) The figure below shows a USGS stream gauge record.



The recurrence interval for a flow of 5000 m3/sec in years is

A. 1.0

B. 1.6

C. 2

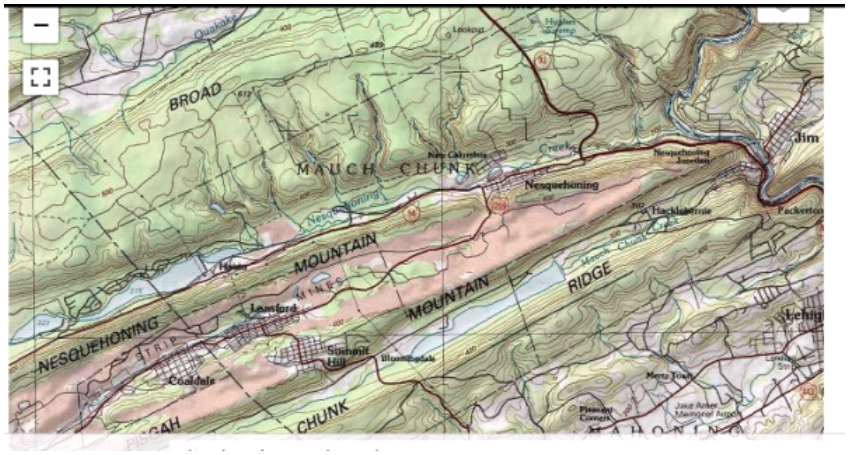
D. 6.0

3. (3 points) Explain how a stream flowing in a channel down a hillside can see its flow increase, even if there are no tributaries joining it and there is no precipitation.

Solution: Key idea is that flow will be fed from groundwater, whether this is simply because downflowing stream intersects an aquifer or because its base drops below the water table.

4. (3 points) A watershed with an area 200 km2 receives 2 cm of rain. What is the average flow in m3/s at the mouth of the watershed if it takes 1 day for the water to drain away?

Solution: 200*1000 m/km*1000 m/km*0.02m /86400 sec=46.9 m3/s (accept answers between 46 and 47 with full credit).



5. (5 points)

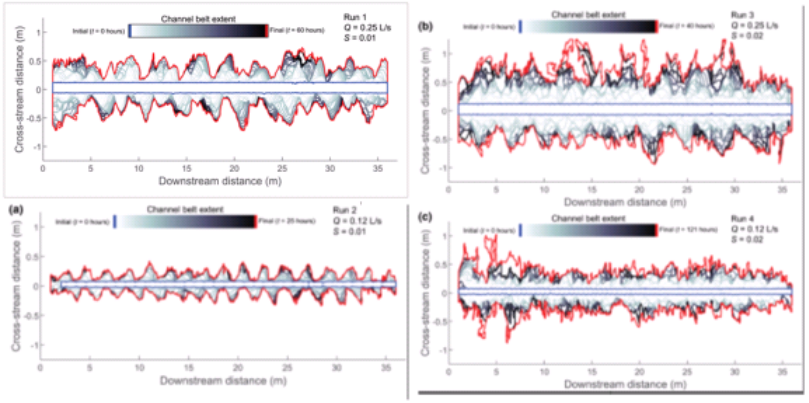
Consider the topographic map at right from the Lehigh Valley in PA.

A. What kind of drainage pattern does this show? (1 point)

B. Mauch Lake is about 2 miles in length. What is the gradient of Nesquethoning Creek between the Lake and where it feeds into the Lehigh River near Jim Thorpe, PA in the northeast (contour interval is 20 feet). (3 points)

C. Are the rivers that run north-south to the north of Nesquehoning Creek in the Mauch Chunk tributaries or distributaries?

Solution: A. Trellis drainage B, 120 feet/4 miles= 120/21000= 0.006 (accept 0.005-0.007) C. Tributaries



6. (2 points)

The figure above shows results from recently conducted experiment that was designed to understand what parameters of a river determine the width of a braided river channel. Understanding this would enable us to use buried river channels to estimate past flows in rivers, but only if we understand what controls the flow.The experiment consisted of a flow on a streamtable, with different possible discharges (Q) and slopes (S). The left hand plots show a slope of 0.01, and the right hand-plots a slope of 0.02. The upper row of plots shows a discharge of 0.25 L/s and the lower row a discharge of 0.12 L/s. The colors show the edge of the bed at various times, with the red lines showing the final extent of the channel belt, after the width is roughly steady with time. (Limaye, Journal of Geophysical Research, 2020)

Which of the following sentences describes the results from these four experiments?

A. Both the average width and size of meanders of the channel bed increase with slope.

B. Both the average width and size of meanders of the channel bed increase with discharge.

C. The average width of the channel increases with discharge, but the size of meanders decreases with discharge.

D. Discharge does not affect the average width of teh channel bed, but slope has a big effect on the average width of the channel bed.

E. No patterns can be discerned from this small set of experiments.



7. (2 points)

The figure above shows the Wind River in Wyoming. Estimate the sinuosity of this river.

A. 1

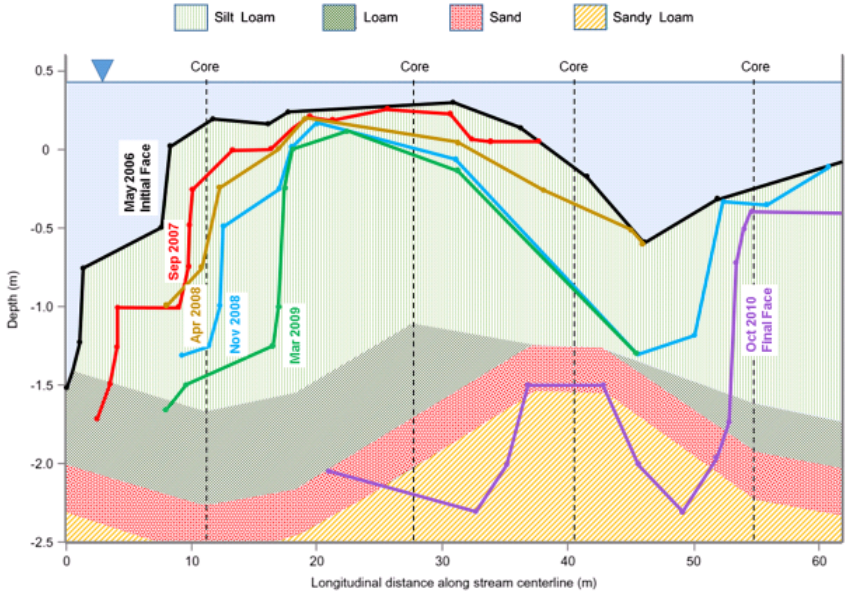
B. 1.6

C. 2.2

D. 2.8

E. 4

F. It is impossible to say from this figure



8. (5 points)

The picture at left shows a crossection of a short section of streambed in central Iowa which was monitored carefully over a five year period. The colored regions show estimates of the different strata in the stream at the beginning of the period while the lines show surveys of the bed taken at different times.

A. Roughly where (give horizontal coordinate) is the knickpoint found in April 2008?

B.How does the knickpoint change over time and why?

C. Between what two periods would you expect to have found the strongest flow and why?

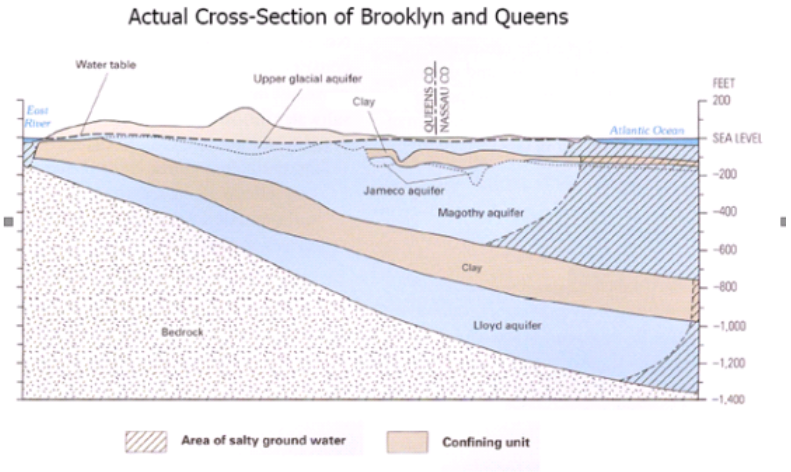
Solution: A. 10m B. Migrates upstream due to erosion. C.March 2009 and October 2010, largest retreat and erosion of heaviest sediment suggests strongest streamflow occurred in this time period

9. (3 points) Explain how faulting can result in stream capture.

Solution: Uplift downstream of the head of the watershed.

10. (2 points) You have a 500 ml beaker full of sediment. Describe how you would estimate the porosity of the sediment.

Solution: Take another beaker with 500 ml of water. Use it to add water to the first beaker until water rises to the top of the sediment. The amount of water removed from the first beaker/500ml is the porosity.



11. (3 points)

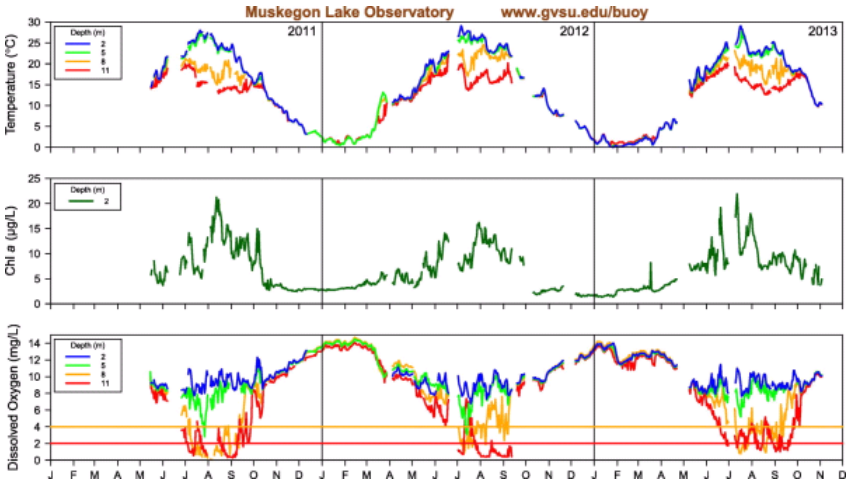
The picture above shows a cross-section across Brooklyn and Queens, NY. Suppose we drill a well to a depth of 1000m on the Nassau County/Queens County Line. Will such a well cause residents of Nassau County who get their water from the Magothy aquifer to experience problems from saltwater intrusion? (1 point) Explain why or why not (2 points).

Solution: No. The well will draw from the Lloyd aquifer, and the presence of a clay layer between the Magothy and Lloyd aquifer will prevent hydraulic gradients from crossing the aquifer boundaries.

12. (2 points) A pressure driven pump can only raise water by 10m, but the Shepherd's tree (Boscia albitrunca), a native of the Kalahari desert can tap groundwater down to a depth of 70m. How is this possible?

Solution: Capillarity- roots and small fissure allow for capillary action to produce very large gradients in hydraulic potential.

13. (3 points)



The plot above shows temperature (top), chlorophyll (middle) and oxygen (bottom) as a function of time in an estuary of Lake Michigan (taken from Biddanda et al., J. of Great Lakes Res., 2018). The different colors show different depths. At oxygen levels below 2 mg/l (62.5 μM) animals that have to breathe oxygen to respire generally die. Explain why the periods of hypoxia found in this estuary occur at the times at which they do.

Solution: Hypoxia is found in the summer, when the lake becomes stratified by temperature and plants bloom more strongly at the surface. As these plants die and the organic matter sinks into the deeper part of the lake it rots and consumes oxygen.

14. (4 points) Describe the processes involved in forming a kettle lake.

Solution: Kettle and kame topography forms as glaciers retreat, with kettles being holes left behind by stranded blocks of ice, which then fill as the water table rises.

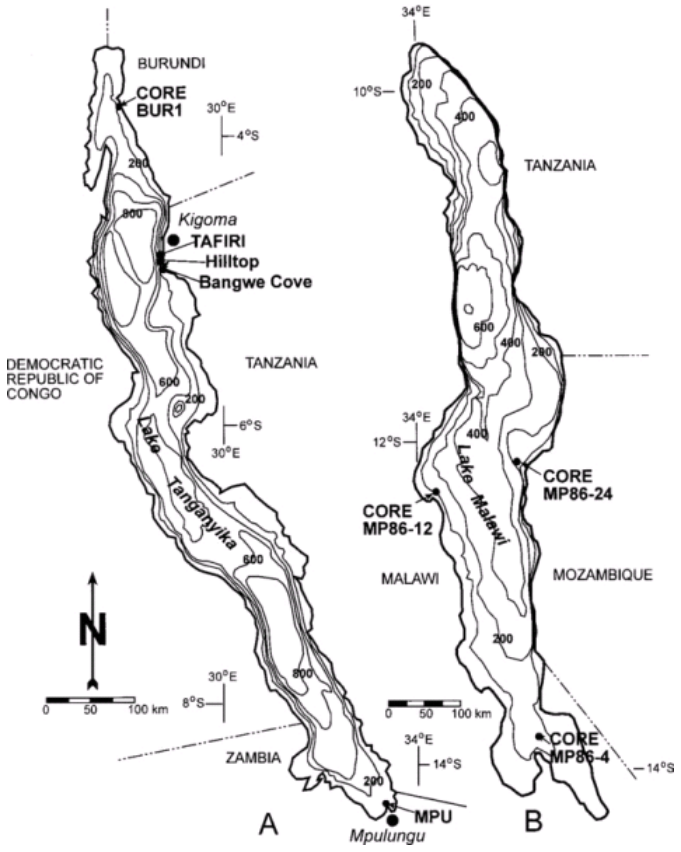
15. (4 points) 

The feature at left is often found in karst terrain.

A. What is it? (1 point)

B. How does it form (2 points) and how does the rock type determine where it is found? (1 points)

Solution: A. Sinkhole B. It forms from the dissolution of rock and is thus most common in carbonate rocks that are easily dissolved by acidic rainfall.



16. (3 points)

The plot at left shows contours of Lake Tanganika (left), which lies to the south of Lake Malawi (right) in the center of Africa. These lakes are examples of grabens.

A. How do we know this? (1 point)

B. What does this mean about how they were formed? (2 points)

Solution: A. Steep sides with relatively flat bottom. B. Part of an opening rift.

17. (5 points) A. Write an equation describing the natural water balance for a soil moisture, with terms that add water as positive terms and terms that remove water as negative terms. (3 points)

B, Describe two ways in which this equation would be modified by human land use (additional terms or changes in the magnitude of individual terms).(2 points)

Solution: A. Change in soil moisture=precipitation-(evaporation+transpiration/evapotranspiration+runoff) B. Add irrigation/note that impermeable surfaces increase runoff/note that agriculture can increase transpiration

18. (4 points) Define the potential evapotranspiration (1 pt) and describe at least three parameters that it depends on (1 point each).

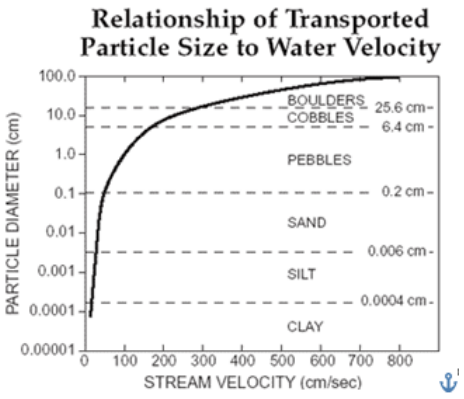
Solution: Potential evapotranspiration is the maximum amount of water vapor that could be evaporated or transpired given the atmospheric conditions (wind speed, air temperature, insolation, relative humidity)

19. (3 points) A lake 250km long and 100 km wide receives 1m of rainfall each year and sees evaporation of 0.6m/yr. If the average inflow to this lake is 1000 m3 /s, what is the average outflow?

Solution: Amount added to lake is 2.5e5*1e5m*0.4m/3.17e7 sec=1e10m^3/$3.17e7sec = 317m3/sec - outflowis1317m3/sec$.

20. (2 points) Which of the following are reasons for removing dams from rivers (mark all that are correct).

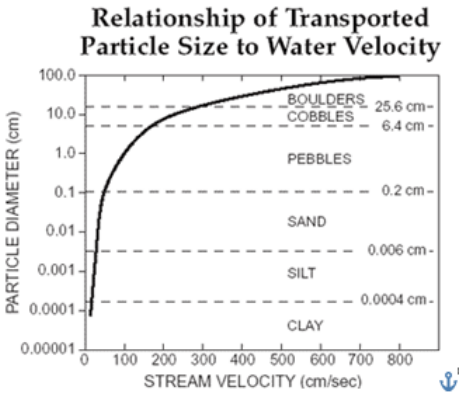
- ◯ Reducing flooding.
- √ **Allowing fish migration.**
- ◯ Reducing the effects of upstream land impacts on stream water quality.
- √ **Filling of the reservoir behind the dam with sediment.**
- ◯ All of the above.
- ◯ None of the above



21. (1 point)

In the plot above cobbles have a particle diameter between ___ cm and ___ cm

6.4 **25.6**



22. (1 point)

During a flood, a stream with a flowing at a speed of 200 cm/s has the ability to transport (mark all correct answers)

- √ **clay**
- √ **silt**
- √ **sand**
- √ **pebbles**
- √ **cobbles**
- ◯ boulders

23. (3 points) Consider the following 3 rivers

River A: Maximum discharge 100 m3/s, width at mouth 10m, depth at mouth 10m

River B: Maximum discharge 1000 m3/s, width at mouth 300m, depth at mouth 10m.

River C: Maximum discharge 100,000 m3/s, 20,000m, depth at mouth 20m.

Which river has the highest sediment competence?

A. A

B. B

C. C

D. B and C

E. It is impossible to say.



24. (1 point)

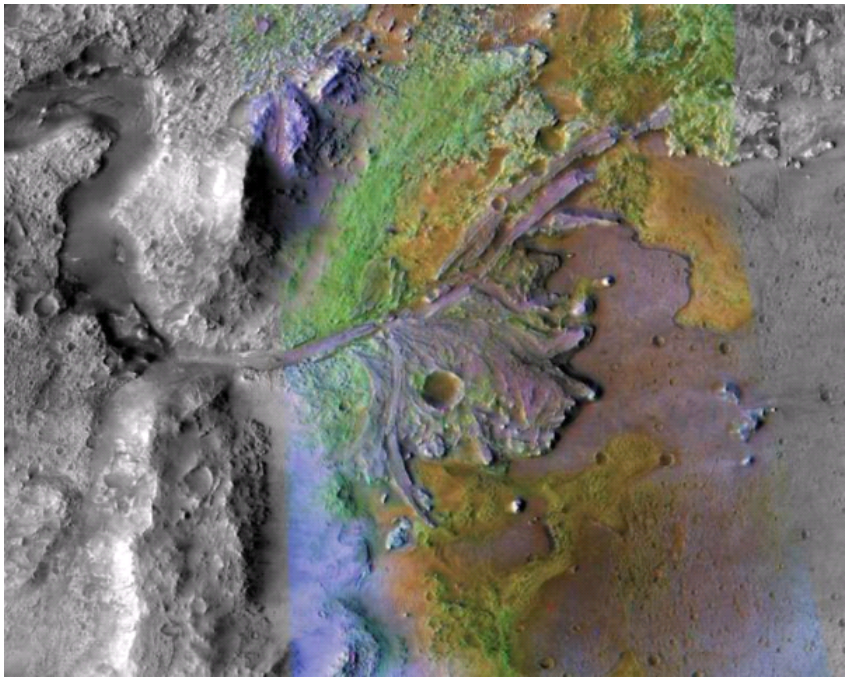
The picture at left shows a

A. Poorly sorted sedimentary deposit with the most likely original flow direction from right to left.

B. Well-sorted sedimentary deposit with the most likely original flow direction from right to left.

C. Poorly sorted sedimentary deposit with the most likely original flow direction from left to right.

D. Well sorted sedimentary deposit with the most likely original flow direction from left to right.

25. (2 points) 

The Curiosity rover on Mars was sent to explore the feature shown in the false color image at left. Given the goal of discovering evidence of past life on Mars why does it make sense to go to this location?

Solution: It is a fossil river delta.

26. (1 point) True of false- A sewage outflow is an example of nonpoint pollution.

    A. True

    B. False

27. (1 point) True or false- Inflow of groundwater contamined with agricultural chemicals to an estuary is an example of nonpoint pollution.

    A. True

    B. False

28. (3 points) The plot at left shows the Patapsco watershed in Maryland. List the stream order for the streams denoted by A, B and C



1	1	2

29. (6 points) Define the following (2 points each)

A. Permafrost

B. Oxbow lake

C. Pingo

Solution: A. Ground that is permanently frozen B. A lake formed by a meandering channel that cuts off a loop. C. A hill with a core of ice, found in permafrost environments.

Conclusion: Thanks for taking this test. I hope you've enjoyed learning about how the water that gives life to us all circulates through our planet.